

Cagayan State University
Carig Campus Tuguegarao City Cagayan
Final Examination in Biochemistry

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Score: _____

Course: BS in Biology- 2C

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I. MULTIPLE CHOICE: Encircle the best answer.

1. Which of the following enzyme is not involved in galactose metabolism?

A. Glucokinase

B. Galactokinase

C. Galactose-1-Phosphate uridyl transferase

D. UDP-Galactose 4-epimerase

2. Which of the following enzyme is defective in galactosemia-a fatal genetic disorder in infants?

A. Glucokinase

B. Galactokinase

C. Galactose-1-Phosphate Uridyl transferase

D. UDP-Galactose 4-epimerase

3. In liver, the accumulation of which of the following metabolite attenuates the inhibitory of ATP on phosphofructokinase?

A. Glucose -6-Phosphate

B. Citrate

C. Fructose-1,6-Biphosphate

D. Fructose -2,6-Biphosphate

4. Mutation in which of the following enzymes leads to a glycogen storage disease known as Tarui's disease?

A. Glucokinase

B. Phosphofructokinase

C. Phosphoglucomutase

D. Pyruvate Kinase

5. Erythrocytes undergo glycolysis for the production of ATP. The deficiency of _____ enzyme leads to hemolytic anemia.

A. Glucokinase

B. Phosphofructokinase

C. Phosphoglucomutase

D. Pyruvate Kinase

6. Cancer cells have high energy demands for replication and division. Increased flux of glucose into glycolysis replenishes the energy demand. Which of the following enzymes plays an important role in tumor metabolism.

A. Glucokinase

B. Phosphofructokinase

C. Phosphoglucomutase

D. Pyruvate kinase M2

7. Which of the following glucose transporters (GLUT) are important in insulin-dependent glucose uptake?

a. GLUT1

b. GLUT2

c. GLUT3

d. GLUT4

8. Which of the following glucose transporters (GLUT) is present in beta cells of the pancreas?

A. GLUT1

B. GLUT2

C. GLUT3

D. GLUT4

9. Which of the following glucose transporter (GLUT) is important in fructose transport in the intestine?

- A. GLUT1
- B. GLUT2
- C. GLUT3
- D. GLUT4

10. Which of the following metabolite negatively regulates pyruvate kinase?

- A. Fructose-1,6-Biphosphate
- B. Citrate
- C. Acetyl CoA
- D. Alanine

11. In the absence of oxygen, pyruvate is converted into lactate in muscle. because

- A. lactate is the substrate from the downstream pathway
- B. Lactate acts as a substrate for the formation of amino acid
- C. During the product of lactate two ATP are produced
- D. During lactate formation, NADH is reconverted into NAD.

12. Which of the following glycolytic enzyme is inhibited by the accumulation of long-chain fatty acid in the liver.

- A. Hexokinase
- B. Glucokinase
- C. Phosphofructokinase
- D. Pyruvate kinase

13. Which of the following statement related to phosphofructokinase -1 is false.

- A. PFK-2 is the isoenzyme of PFK -1 that is present in the liver
- B. PFK-1 is activated by AMP whereas inhibited by ATP and citrate
- C. The binding of ATP to PFK -1 induces the conformation change from R to T state.

D. PFK-1 is regulated by post translational modification such as phosphorylation.

14. Which of the following statement about Phosphofructokinase -2 (PFK-2) is false?

A. PFK-2 is a bifunctional enzyme having a kinase domain, phosphatase domain, and a regulatory domain

B. Activated protein kinase A phosphorylates PFK-2 and activates phosphatase domain

C. PFK-2 catalyzes the conversion of fructose -6 -phosphate to fructose 2-6 biphosphate

D. PFK-2 phosphatase activity is activated by the insulin signaling pathway

15. Which of the following hormone decreases blood glucose and increases the uptake of glucose in various tissues like skeletal muscle, adipose tissues?

A. Glucagon

B. Epinephrine

C. Cortisol

D. Insulin

16. Which of the following statement is True?

A. Glycolysis occurs only in mammalian cells

B. Glycolysis occurs in mitochondria

C. Glycolysis occurs in the presence and absence of oxygen

D. Glycolysis occurs when ATP concentration is high

17. What is the rate-limiting step in Glycolysis?

A. Hexokinase

B. Phosphohexose isomerase

C. Glyceraldehyde-3-phosphate dehydrogenase

D. Enolase

18. The net gain of ATP during the conversion of glucose to pyruvate is :

A.1 ATP

B.2 ATP

C.1 ATP +1 GTP

D.4 ATP

19. During the conversion of glucose to pyruvate, two NADH molecules are generated. Which of the following steps generates NADH?

- A. Conversion of Fructose-6 phosphate to fructose -1,6-bisphosphate
- B. Conversion of glyceraldehydes-3-phosphate to 1,3-bisphosphoglycerate
- C. Conversion of 3-phosphoglycerate to 2-phosphoglycerate
- D. Conversion of phosphoenolpyruvate to pyruvate

20. What is the committed step in Glycolysis?

- A. Conversion of glucose to glucose -6-Phosphate
- B. Conversion of fructose -6-phosphate to fructose -1,6-bisphosphate
- C. Conversion of glyceraldehyde-3-phosphate to 1,3-bisphosphoglycerate
- D. Conversion of 3-phosphoglycerate to 2-phosphoglycerate

21. Glycolysis consists of three irreversible steps. Which of the following enzyme-catalyzed reactions are not irreversible steps in Glycolysis?

- A. Hexokinase
- B. Phosphofructokinase
- C. Glyceraldehyde-3-phosphate kinase
- D. Pyruvate kinase

22. The following are the negative regulators of phosphofructokinase except?

- A. ATP
- B. AMP
- C. Citrate
- D. pH

23. Which of the following steps is inhibited by sodium fluoride?

- A. Glyceraldehyde -3-phosphate dehydrogenase

B. 3-phosphoglycerate mutase

C. Enolase

D. AMP

24. Which of the following step is inhibited during arsenate poisoning?

A. Glyceraldehyde-3-phosphate dehydrogenase

B. 3-phosphoglycerate mutase

C. Enolase

D. Pyruvate kinase

25. Which of the following enzyme catalyzes the conversion of pyruvate to lactate?

A. Pyruvate reductase

B. Lactate reductase

C. Lactate dehydrogenase

D. Pyruvate dehydrogenase

bonded to it.

26. Gluconeogenesis is the production of glucose from non carbohydrate molecules. Which of the following is not substrate for Gluconeogenesis?

A. Lactate

B. Alanine

C. Glycerol

D. Acetyl CoA

27. Gluconeogenesis occurs in the liver and kidneys. Which of the following enzyme are important for gluconeogenesis are expressed exclusively in these tissues?

A. Glucose -6-phosphatase

B. Fructose-1,6-Biphosphatase

C. Phosphoenolpyruvate carboxykinase

D. Pyruvate Carboxylase

28. During gluconeogenesis, the three irreversible steps of glycolysis have to be bypassed. The first step is the conversion of pyruvate to phosphoenolpyruvate. Which of the following statements is false regarding the reaction step?

A. This reaction involves a two-step process catalyzed by pyruvate carboxylase and phosphoenolpyruvate carboxykinase

B. Conversion of oxaloacetate from pyruvate occurs in mitochondria and is shuttled into the cytosol.

C. Formation of phosphoenolpyruvate requires both ATP and GTP as an energy source.

D. Acetyl CoA is an activator of the enzyme pyruvate carboxylase.

29. During gluconeogenesis, the three irreversible steps in Glycolysis have to be bypassed. The final step is the conversion of Glucose-6-phosphate to glucose-6-phosphatase. Which of the following statements is true about the reaction step?

A. Conversion of glucose-6-phosphate to glucose releases one ATP molecule.

B. It is a highly active enzyme in skeletal muscle

C. Defect in glucose-6-phosphatase leads to abnormal accumulation of glycogen in the liver

D. The reaction occurs in mitochondria

30. During prolonged starvation, which of the following hormones is responsible for increasing gluconeogenesis in the liver.

A. Insulin

B. Glucagon

C. TSH

D. Thyroxine

31. Regulated metabolic pathways are

A. Irreversible

B. Committed after the first step

C. Usually regulated in the first step

D. All of the above are correct.

32. Each reaction in a metabolic pathway is :

A. Controlled by the end product

B.Catalyzed by a specific enzyme

C.irreversible

D.reversible

33.The hormones ,glucagon and epinephrine,stimulate glycogen breakdown to G-6-P

A.directly by binding to glycogen phosphorylase

B.indirectly by first stimulating adenylate cyclase to make cAMP

C.only in the liver

D.only in the muscles

34.Activation of the _____kinase results in the activation of _____kinase and thereby the phosphorylation of both glycogen phosphorylase and glycogen synthase.

A.pyruvate phosphorylase

B.phosphorylase,cAMP dependent protein

C.cAMP dependent protein phosphorylase

D.cAMP dependent protein pyruvate

35.The major free energy sources for anabolic pathways are:

A.ATP and NADPH

B.ATP and NADP

C.ADP and NADPH

D.ADP and NADP

II.MODIFIED TRUE OR FALSE: Write TRUE if the statement is correct and FALSE if its incorrect.Underline the word /words that makes the statement incorrect.(2 points each)

_____TRUE_____1.During strenuous exercise,conditions in muscle cells change from aerobic to anaerobic as the oxygen supply becomes inadequate to meet the demand.Such conditions cause pyruvate to be converted to lactate rather than acetyl CoA.

_____FALSE_____2.A decreased amount of carbonic acid causes blood pH to rise,which makes the blood slightly more basic.

_____TRUE_____3.Lactate accumulation can also occur in heart muscle if it experiences decreased oxygen supply .

____FALSE____ 4. Glycogen is a branched polymeric form of glucose, is the storage form of carbohydrates in humans and animals.

____TRUE____ 5. Glycogenesis is the metabolic pathway by which glycogen is synthesized from glucose -6-phosphate.

____TRUE____ 6. Gluconeogenesis is the metabolic pathway by which glucose is synthesized from non carbohydrate materials.

____TRUE____ 7. The process of gluconeogenesis and glycolysis are not exact opposites.

____TRUE____ 8. Glycolysis is the only pathway by which glucose may be degraded.

____TRUE____ 9. The pentose phosphate pathway metabolites glucose to produce ribose (a pentose), NADPH, and other sugars needed for biosynthesis.

____FALSE____ 10. Phospholipase A1 is an enzyme present in the venom of bees and vipersnakes.

____TRUE____ 11. Lipase is an enzyme that specifically digests butter fat in our food.

____TRUE____ 12. Triglycerides are the main constituents of body fat in humans and other animals as well as vegetable fat.

____TRUE____ 13. Triglycerides are major components of very low density lipoprotein (VLDL).

____TRUE____ 14. High fat foods remain in the stomach longer than low fat foods.

____TRUE____ 15. Cholesterol is abundant in cell membranes.

III. ESSAY. Answer the following questions briefly. (5 points)

1. What chemical condition is responsible for jaundice?

- Jaundice is known as hyperbilirubinaemia. It is caused by too much bilirubin in the blood.

2. Which bile pigment is responsible for the yellow color of urine?

- Normal urine color ranges from pale yellow to deep amber. The result of a pigment is called Urobilin together with the urochrome and how diluted or concentrated the urine is.

3. What is the general name for the building blocks from which a DNA molecule is made?

- DNA molecule is made up of chemical building blocks called Nucleotides

4.What are the names of the pentose sugar present present ,respectively , in DNA and RNA

Molecules?

- In DNA, the five carbon sugar is called deoxyribose that is found in this type of polynucleotide and in RNA, the sugar is called ribose.

5.What type of molecule is the starting material for the transcription of phase of protein synthesis?

- The RNA polymerase or the RNA pol. It is when this starts to attaches to the template of DNA strands and begin to catalyzed the production of complementary RNA.

6.How does recombinant DNA differ from normal DNA?

- Recombinant DNA differs from the normal DNA in that former results from artificial methods in the test tube, whilst the latter is a normal biological process that results in the remixing of existing DNA sequences in essentially all organisms.

7.What is the difference between a cofactor and a coenzyme?

- Cofactor doesn't have the ability to bind itself to enzyme, while coenzyme is quite often loosely bind to the active site of an enzyme and aid in substrate recruitment.

8.All coenzymes are cofactors,but not all cofactors are coenzymes.Explain this statement.

- Both are small organic molecules, the only difference is that cofactors functions in the enzyme activity only while coenzymes serves as a cofactor and assists in biological transformations. Hence all coenzymes are cofactors, but not all cofactors are coenzymes.

9.Briefly explain how carbon atoms from amino acids can end in ketone bodies.

- Most of the carbons from amino acid degradation are converted to pyruvate, intermediates of the TCA cycle or acetyl CoA. Thus, during fasting these carbons are converted into glucose to the ketone bodies such as liver and kidney. Perhaps, they can be used for lipogenesis in a well fed state.

10.How are the amino acids from protein "processed" when an individual is in a state of starvation?

- In time of starvation, the amino acids can be used as an energy source and processed through the Krebs cycle. Also, amino acids become the major source for glucose homeostasis because of the decrease in plasma insulin level and rise in glucocorticoid level.

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